

Where fatality enters the record: an outcome-observability audit of insurer, transit-authority and prosecutor crash data in twelve Mexican cities

Alternative titles. (a) Collisions are recorded, outcomes are not: fatality observability in Mexican urban crash registries. (b) The outcome field: how insurer, transit-authority and prosecutor registries differ in recording road fatalities across twelve Mexican cities.

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Outline, not a draft: title and abstract are written out; the remaining sections are given as bullets.

ABSTRACT

Mexican crash registries record collisions well and outcomes badly. The collision and its fatal outcome are captured by different institutions under different mandates, and the point at which fatality enters the record has never been measured. We audit three administrative registries across twelve cities of two states, 2019–2023: a private insurer (AXA), the national transit registry (ATUS/INEGI), and prosecutor case files obtained through freedom-of-information requests. The insurer records 18 deaths across 75,962 crashes, the transit registry 914 across 270,533 events, and prosecutors open 2,450 fatal case files over the same cities and years. To test whether the missing outcome can be recovered from what is recorded, records are aggregated on an H3 hexagonal grid and each collision layer is scored on how much of the fatal outcome it locates, against the maximum any selection of the same number of cells could reach. A threshold-free summary of the full curve places the insurer at 0.583 and the transit registry at 0.563, where 0 matches the optimal ordering and 1 matches random selection — and, decisively, the same holds when the prosecutor’s own injury case files are used to locate its own fatal case files (0.602), a comparison that crosses no institutional boundary. Since no institutional boundary is crossed in that last comparison, the shortfall is not a consequence of fragmentation and would not be repaired by integrating the registries: in every registry examined, collision location does not carry fatality location. The same layers perform markedly better against pedestrian and cyclist casualties (0.503 against a 0.391 within-registry floor), separating two objectives that municipalities currently plan from one input. A monotone motorcycle-involvement gradient across the three sources (0.5–0.7%, 6.8–8.8%, 17.0%) locates where in the institutional chain severity becomes visible, and a two-source capture–recapture in the state that released event timestamps places fatal events at no fewer than 1,900 against 594 held by the transit registry. Fatality is therefore a recording problem rather than an analytical one, and the remedies belong at the recording layer. The prosecutor layer is corroborated externally: INEGI mortality registrations, produced through the civil-registry chain, record 1,508 and 1,864 transport deaths in the two state groups, placing the prosecutor at 69–81% of the mortality registry, the transit registry at 22–32% and the insurer below 1% — and, in Nuevo León, landing within 2% of the capture–recapture floor derived independently from record matching. Records are clipped to the official urban polygon rather than to a hand-drawn boundary, and what falls outside is retained as a periurban stratum where the transit registry logs 424 events and 9 deaths while the prosecutor opens 229 fatal case files — a seventeen-fold concentration of lethality at the urban edge, in the band the transit registry barely sees. We document a release asymmetry between two neighboring prosecutors — one reserved coordinates, the other victim attributes — that decides which analyses each state permits.

1. Introduction

- The object of study is the outcome field, not the map. A collision record and its fatal outcome are produced by different institutions: insurers close property-damage claims, the transit registry counts events, prosecutors open a case file once someone dies. Nobody is responsible for carrying the outcome back to the collision record.

- Consequence in numbers, stated up front: across the same twelve cities and five years, the insurer records 18 deaths, the transit registry 914, and prosecutors open 2,450 fatal case files.
- The usual response is analytical — infer severity from collision density, or from exposure models. We test whether that inference is available at all, and the test is spatial because location is the one attribute all three registries share.
- Gap addressed: existing data-quality work measures underreporting against a presumed true count. It does not ask whether the outcome is recoverable from what is recorded, which is the question a city actually faces.
- Contribution, stated without novelty inflation: (i) a substitutability test — can collision location stand in for fatality location — with a budget-normalized ceiling and a threshold-free summary; (ii) a within-registry control that separates structural from institutional causes; (iii) a georeferenced prosecutor layer for twelve cities, and a documented account of what each authority released and reserved.

2. Related work

- Hotspot identification and spatial prioritization: the empirical-Bayes and network-screening tradition; the H3/hexagonal grid literature; MAUP.
- Crash underreporting and record linkage: police–EMS–trauma linkage studies; capture–recapture in injury surveillance.
- Closest antecedents to cite explicitly: hotspot-localization comparisons across data sources in Rwanda and Sri Lanka.
- **BLOQUEANTE** **Prior Mexican uses of prosecutor / FGJ crash data must be located and cited.** This is not the first such use and the manuscript must not imply it. Search INEGI–FGJ linkage work, state observatory reports (CDMX, Jalisco, Nuevo León) and the Mexican injury-epidemiology literature; the claim we can defend is the *comparative* audit across three registries and twelve cities, not first use of the source.
- **IMPORTANT** Differentiation paragraph against the authors' own manuscript under review (IATSS Research, five-pillar observability audit, Mexico City). That paper audits *one* source field by field and asserts that each severity level connects to an institutional mandate; this paper is the empirical test of that assertion across sources and cities. Naming it pre-empts a fragmentation objection.
- Positioning sentence: the contribution is a reframe plus new data, not a new estimator, and not a new source — say so rather than let a reviewer say it.

3. Data and study area

3.1 Cities

- Twelve municipalities drawn from a national top-100 list: six in Tamaulipas (Tampico, Ciudad Madero, Reynosa, Victoria, Nuevo Laredo, Matamoros) and six in Nuevo León (Monterrey, San Nicolás, Apodaca, Escobedo, Santa Catarina, García).
- Spatial analysis runs on Tamaulipas only, because Nuevo León's prosecutor reserved coordinates. Nuevo León supplies counts and victim attributes.
- **IMPORTANT** Guadalupe (top-100) and San Pedro Garza García (VRU list) are the subject of a supplementary request; integrate on arrival, if necessary at revision stage.
- Main text carries two spatial cases — **Tampico** (anchor) and **Reynosa** (contrast); the other four move to supplementary material to keep the results readable.

3.2 The three sources

SOURCE	RECORDS	DEATHS	UNIT	GEO	TIME	VICTIM ATTRIBUTES
AXA / I2DS (insurer)	75,962	18	claim	point	date + hour	vehicle type only
ATUS / INEGI (transit)	270,533	914	event	point	date + hour	role, sex, age
Prosecutor — Tamaulipas	1,222 carpetas	1,222	case file	point	filing date only	reserved
Prosecutor — Nuevo León	1,228 carpetas	1,279 victims	case file	reserved	date + hour	sex, age band, vehicle
INEGI mortality (EDR) reference	3,372	3,372	death	municipality	date of death	sex, age

Table 1. Units are never summed across columns. The common unit across both prosecutors is the case file, giving a headline of 2,450 fatal-crash case files; victims are reported only where released.

- **Prosecutor counts validated against INEGI mortality microdata** (EDR 2019–2024 pooled, filtered by year of *occurrence*, ICD-10 V01–V89). Deaths are counted through an institutionally independent chain — death certificate, civil registry, INEGI — so agreement is genuine external corroboration. Because `mun_ocurr` records where the death occurred, often a hospital, the comparison is made by metropolitan group and municipality-by-municipality only for the four isolated cities. See §5.6.
- Data-release asymmetry as a governance finding: Tamaulipas released geography without attributes; Nuevo León released attributes without geography (CNPP arts. 105, 106, 218). Each reservation eliminates a different class of analysis.
- Brief four-state governance paragraph (adding Querétaro C5 and Chihuahua FGE) — no maps, no city-level data. The full governance treatment is reserved for a separate manuscript with more states.

4. Methods

4.1 Grid, clipping and strata

- H3 hexagons at resolution 9 (~0.12 km², block scale) and 10 (~0.017 km², corner scale). Resolution 9 carries the headline; at resolution 10 the ceiling saturates at 100% and the figure is reported as an upper bound only.
- **The study area is the official urban polygon, not a hand-drawn boundary.** Urban AGEBs from INEGI's 2020 Marco Geoestadístico are dissolved per municipality with a 150 m allowance, since AGEBs terminate at the centerline of the perimeter road and a crash on that road would otherwise fall outside. Resulting areas range from 53 km² (Ciudad Madero) to 195 km² (Reynosa).
- **Nothing is discarded.** Records the municipality reports but that fall outside the polygon form a *periurban* stratum, reported in its own right (§5.5). This resolves the boundary problem rather than hiding it: any correct urban boundary removes more prosecutor records than transit-registry records, because lethality concentrates at the edge, so a clip that discards would bias the comparison in favor of whichever registry is most urban.
- **MENOR** The AGEB layer is 2020 vintage, so peripheral growth during 2021–2023 sits outside the urban stratum and enters the periurban one.
- Every record's stratum assignment is published with the replication package, so the clip is auditable rather than asserted.

4.2 The metric

- **Coverage.** Rank cells by the audited frequency layer, take the top $b\%$, and report the share of the outcome falling inside.
- **Ceiling.** The same share when cells are ranked by the outcome itself — the maximum any selection of the identical number of cells could achieve. Normalizing this way keeps the comparison fair; the cell budget is an instrument for measuring substitutability, not a claim about municipal finance.
- **ABC.** Area between the observed and optimal coverage curves across all budgets, normalized by the area between optimal and the random diagonal. 0 = matches the optimal ordering; 1 = matches random selection. Removes the arbitrary threshold entirely.
- **BLOQUEANTE** **Tie-breaking must be specified.** 80–97% of cells share a frequency value, so the order within tied blocks drives the metric — up to 0.20 either way. We use the exact expectation under random tie-breaking: in a block of m tied cells summing S , taking j captures $(j/m) \cdot S$ on average. Pessimistic and optimistic bounds are reported alongside.
- Spearman rank agreement is reported as a secondary descriptive only, with the tie problem stated — the same excess-ties issue makes it a weak instrument here.

4.3 Audit design: layers × objectives

- Two audited layers (what a city would use to prioritize): insurer frequency, transit-registry frequency.
- Two policy objectives (what the city is trying to prevent): fatalities (prosecutor homicide case files) and pedestrian + cyclist casualties (transit registry).
- **Structural-floor run.** Within a single registry, order cells by that registry's *non-target* events and predict its target events. This isolates how much of the shortfall is intrinsic to using frequency as a proxy for severity, with institutional difference held out.

- **IMPORTANT** The floor run must exclude the outcome from the frequency layer. Leaving homicide case files inside the prosecutor's own frequency layer produces 0.134 instead of 0.641 — the layer contains its own target and the result is an artifact. State the exclusion explicitly in Methods.

5. Results

5.1 Collision location does not carry fatality location

OBJECTIVE	INSURER	TRANSIT REGISTRY	STRUCTURAL FLOOR
Fatalities (prosecutor)	0.583	0.563	0.602
Pedestrian + cyclist casualties	0.503	—	0.391

Table 2. ABC, resolution 9, urban stratum, median of six cities. Lower is better; 1.00 equals random selection. The structural floor orders each registry against itself, with the target excluded from the frequency layer.

- The official transit registry does not outperform the private insurer. The finding is about the recording regime, not about one provider.
- The prosecutor's own injury case files locate its own fatal case files no better (0.602) — in fact marginally worse than either external registry. Since no institutional boundary is crossed there, the shortfall cannot be attributed to fragmentation, and integrating the registries would not close it.
- Interpretation to state plainly: within every registry examined, where crashes accumulate is not where people die.
- Against pedestrian and cyclist casualties both layers improve markedly, and there the floor (0.391) sits clearly below the cross-source result (0.503), so a real if modest institutional penalty does exist for that objective.
- Operational consequence: fatality reduction and VRU protection are two programs, currently planned from one map. The existing maps serve the second and not the first.

5.2 The conclusion does not depend on the 10% threshold

BUDGET	COVERAGE OBSERVED	CEILING
5%	12.3 – 24.6%	41.7 – 60.4%
10%	23.4 – 50.0%	60.4 – 80.0%
15%	30.0 – 52.9%	78.6 – 100%
20%	42.3 – 68.2%	94.8 – 100%

Table 3. Insurer layer against prosecutor fatalities, resolution 9, urban stratum, range across six cities.

- The gap persists at every budget and in every city; 10% was arbitrary, not favorable. ABC is the primary indicator and the table is the illustration.
- Scale sensitivity: at resolution 10 the ABC median rises to 0.951 and two cities exceed 1.00 (Reynosa 1.079, Victoria 1.007). Phrase carefully — the head of the ranking retains signal in the operational 5–20% band, while the remainder of the ordering stops informing. Not a flat "worse than random."

5.3 The mechanism: a motorcycle gradient across sources

- Motorcycle involvement rises monotonically with institutional distance from the insurance claim: 0.5–0.7% (insurer), 6.8–8.8% (transit registry), 17.0% (Nuevo León prosecutor).

- In Tamaulipas, casualties in motorcycle-involved crashes (1,483) exceed pedestrian plus cyclist casualties (1,165) — motorcycles warrant their own layer, not absorption into a VRU aggregate.
- Institutional explanation, not an accusation: an insurer file exists to close a property-damage claim, and a fatality moves the case out of the patrimonial channel into the criminal one. The recording stops for a structural reason. This is the strongest available version of the argument and it yields a cheap recommendation — one mandatory outcome field, fed from the police report rather than the adjuster.
- **MENOR** Measurement caveat: the transit registry flags motorcycle involvement at the event but does not attribute the casualty to a vehicle, so the layer is defined as occupant casualties in motorcycle-involved crashes excluding pedestrians and cyclists, and overstates.

5.4 How much is missing: capture-recapture in Nuevo León

- Matching fatal transit-registry events (557) to prosecutor homicide case files (1,228) on municipality + same day + ± 1 hour yields 356 matches; a date-shift null yields 11.6 by chance — a $30.7\times$ excess confirming the linkage is real, not coincidental.
- Lincoln–Petersen gives at least 1,900 fatal events; the transit registry would see $\leq 29\%$ and the prosecutor $\leq 64\%$. Positive dependence between the sources means the estimator understates, so 1,900 is a floor, not a point estimate.
- Not feasible in Tamaulipas: that prosecutor released the case-file opening date rather than the event date, leaving only space + year, which gives 55 matches against 12.8 by chance and an implausible estimate. A single reserved field eliminates an entire class of analysis — that is the governance argument, made with a number.
- **External confirmation.** The estimate is built only from matching transit-registry events to prosecutor case files. Counted independently through the civil-registry chain, INEGI records 1,864 transport deaths in the same six municipalities and years — within 2% of the 1,900 floor. The capture–recapture is therefore corroborated rather than assumed.
- Path to a defensible estimate, for the journal extension: INEGI mortality registrations as a third source and a log-linear model with interaction terms, so independence is not assumed; Chao's lower bound as the conservative fallback.

5.5 The periurban stratum: where lethality concentrates and the registry goes blind

STRATUM	INSURER	TRANSIT EVENTS	TRANSIT DEATHS	PROSECUTOR FATAL FILES	FATAL FILES PER 1,000 TRANSIT EVENTS
Urban (AGEB)	26,176	36,476	322	1,143	31
Periurban	1,913	424	9	229	540

Table 4. Six Tamaulipas cities, 2019–2023, by stratum.

- Outside the urban polygon but inside the municipality, the transit registry logs 424 events and 9 deaths while the prosecutor opens 229 fatal case files — 25 prosecutor fatalities for every death the transit registry sees there, against 3.5 inside the city.
- Normalized by event volume the concentration is seventeen-fold: 540 fatal case files per 1,000 transit events in the periurban band versus 31 in the urban one. The urban fabric is where the official registry works; the edge is where it stops working, and it stops precisely where people are killed.

- **MENOR** Report counts, not ratios, for Tampico and Ciudad Madero: the transit registry logs 2 and 0 periurban events respectively, so a per-event ratio there has no denominator to stand on. The raw contrast carries the point (Tampico: 0 transit deaths against 13 prosecutor fatal files).
- City-level ratios with adequate denominators: Matamoros 22.8×, Victoria 19.9×, Reynosa 15.9×, Nuevo Laredo 8.0×.

5.6 External validation against the mortality registry

GROUP	INEGI DEATHS	PROSECUTOR	TRANSIT REGISTRY	INSURER
Tamaulipas, six cities	1,508	1,222 81.0%	331 21.9%	12 0.8%
Nuevo León, six municipalities	1,864	1,279 68.6%	594 31.9%	6 0.3%

Table 5. Transport deaths (V01–V89) occurring 2019–2023, against each registry's own count for the same municipalities. Percentages are of the INEGI figure.

- The prosecutor layer is corroborated: it undercounts the mortality registry by a consistent margin (69–81%) rather than diverging, which is what a case-file threshold predicts — a fatal crash must be opened as a traffic homicide, some are classified otherwise, and some deaths follow the file's closure.
- The ladder is monotone and steep. The insurer holds under 1% of the deaths recorded by the civil registry; the national transit registry between a fifth and a third; the prosecutor roughly three quarters. This is the outcome-observability gradient stated in a single table.
- **IMPORTANT** Cause coding differs sharply between the two states and constrains the case definition. Nuevo León codes 1,817 of its 1,864 deaths as V89 (motor-vehicle accident, type unspecified); Tamaulipas distributes across V09 (410), V89 (398), V87 (199), V49 (110) and V29 (92). A V01–V79 parallel therefore collapses in Nuevo León (140 of 1,864) and cannot be reported; V01–V89 is the case definition. The divergence is itself evidence for the manuscript's thesis — two neighboring states under one federal standard record the same event class at different resolutions.
- Sizing the pending Nuevo León request with the same source: Guadalupe accounts for 112 transport deaths in the period and San Pedro Garza García 29.
- **MENOR** Data note for reproducibility: in the 2019 EDR file the key fields carry a trailing tab (28\t), so an unstripped filter returns zero rows for that year silently.

6. Discussion

- The mortality registry supplies the denominator the other three lack, and confirms the ordering is not an artifact of any single institution: under 1% of deaths reach the insurer, a fifth to a third the transit registry, roughly three quarters the prosecutor.
- The failure is at the recording layer, not the analysis layer. No spatial method applied to collision records recovers fatality location, because the information was never written down — the within-registry control is what establishes this.
- What the audit licenses a city to do with the data it already holds: target pedestrian and cyclist protection. What it does not license: siting fatality-reduction interventions.
- Remedies stated where they belong. One mandatory outcome field in insurer records, fed from the police report rather than the adjuster. Event date and hour released by prosecutors. Routine return of the judicial outcome to the transit registry that holds the collision.

- Why integration is not the answer here — the structural-floor result forecloses the intuitive recommendation, which sharpens the policy message rather than weakening it.
- Transferability: the substitutability test needs only a collision layer, an outcome layer and a grid, so it runs wherever two registries coexist. The motorcycle gradient is the portable diagnostic for whether a registry is severity-blind.
- Governance: two neighboring states under the same federal framework released complementary halves of the same record. What a city can analyze is decided by a reservation criterion, not by whether the data exist.

7. Limitations

- The mortality registry counts deaths by municipality of *death*, not of crash, so validation is only valid at the metropolitan-group level; and it covers the whole municipality, so it spans both strata.
- Cause-of-death coding resolution differs between the two states, which constrains the case definition to V01–V89 and prevents a narrower road-traffic parallel.
- Tamaulipas victim counts unavailable, so the victims-per-case-file ratio observed in Nuevo León (1.0415; 1.034–1.058 by year, 1.018–1.088 by municipality) is a declared assumption there and supports no headline figure.
- Insurer coverage varies with insurance penetration, which is not uniform across the twelve cities.
- Transit-registry coverage gaps: Ciudad Madero absent before 2021 and nearly absent in 2023; insurer file gaps in October 2021 and November 2023.
- Capture–recapture assumes closure and imperfect independence; the estimate is presented as a lower bound only.

8. Conclusions

- Collision location does not carry fatality location in any of the three registries audited, including within a single registry — the limit is structural, not a fragmentation artifact.
- The same records do support vulnerable-road-user targeting, which separates two objectives currently planned from one input.
- The motorcycle gradient across sources shows severity becoming visible only where an institutional mandate forces it, which is where the fix belongs.
- Recommendation set, all at the recording layer: a mandatory outcome field in insurer records; event date and hour released by prosecutors, since one reserved field forecloses capture–recapture entirely; and a defined channel returning the fatal outcome to the registry that holds the collision.

Figures and tables

- **Fig. 1** — Coverage-versus-ceiling curves for Tampico and Reynosa, with the 5–20% operational band shaded.
- **Fig. 2** — ABC by city and layer, both objectives, with the structural floor marked as a reference line.
- **Fig. 3** — Motorcycle-involvement gradient across the three sources.
- **Fig. 4** — Divergence map over the urban polygon: insurer frequency against prosecutor fatalities, Tampico and Reynosa, resolution 9, with the periurban band shown outside the boundary.
- **Table 1** — Source comparison (units, geography, time, attributes).
- **Table 2** — ABC by layer and objective, with the structural floor.

- **Table 3** — Budget sensitivity, 5/10/15/20%.
- **Table 4** — Urban versus periurban stratum.
- **Table 6** — Four-state data governance: what each authority released and what it reserved. **IMPORTANT** still to be built.
- **Table 5** — External validation against INEGI mortality, by state group.
- **Fig. 5** — Outcome-observability ladder: share of INEGI-recorded deaths held by each registry.

Open items before submission

- **BLOQUEANTE** Locate and cite prior Mexican uses of prosecutor crash data; remove any first-use implication.
- **IMPORTANT** Supplementary Nuevo León request — Guadalupe (112 deaths in period) and San Pedro Garza García (29).
- **IMPORTANT** Related-work differentiation paragraph against the IATSS manuscript.
- **IMPORTANT** Table 6 (four-state governance).

Outline generated 10 Sep 2026 · figures at H3 resolution 9 unless stated · urban stratum (INEGI AGEb polygon) · all coverage and ABC values computed under expected random tie-breaking